Model Selection Stabilization for Relative-Periodic Decomposition of Cellular Automaton Spacetimes

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Abstract

We prove that Bernoulli NML model selection over relative-periodic backgrounds of cellular automaton (CA) spacetimes stabilizes as the observation window grows: every candidate whose asymptotic per-site NLL rate exceeds the minimum is eventually excluded (Theorem 3). When the rate-minimizing candidate is unique—which holds in all cases we test—the selected model locks to a single winner. The argument rests on a dimension-agnostic *orbit-class reduction*: fitting a relative-periodic background to a binary spacetime decomposes into independent majority voting on orbit classes (Theorem 1), partition refinement along constant-velocity chains makes higher periods monotonically more expressive (Theorem 2), and the Bernoulli NML criterion's $O(\log T)$ complexity penalty is eventually dominated by $\Theta(T)$ data-fit gaps between candidates with distinct rates. We also show by explicit construction that background period recovery is impossible in general without additional assumptions (Theorem 4). The theory applies identically to 1D, 2D, and 3D automata.

Cross-dimensional experiments are consistent with the stabilization prediction: selected periods lock for 1D elementary CA (rules 30, 54, 110), 2D totalistic rules (773 non-trivial from 1,050 candidates, plus all 106 named Life-like rules from the LifeWiki), and a 3D totalistic rule—with margins growing after stabilization. The LifeWiki survey supports the practical relevance of the predicted finite-sample trade-off: Diamoeba transitions from period 1 to period 2 between $T{=}100$ and $T{=}400$ as the NLL gap crosses the complexity threshold, while Fredkin (B1357/S02468) is the only rule achieving period 8. As an application, the framework identifies 2D rules with persistent structured projection residuals, including one (S37/B11) exhibiting extensive residual scaling verified at 400 steps, across multiple seeds, and at grid sizes up to 192×192.

Introduction

Cellular automata (CA) generate complex spatiotemporal patterns from simple local rules. Decomposing these patterns into regular and irregular components is a central problem in CA theory, with applications to understanding emergent computation, classifying rule complexity, and identifying coherent structures.

The computational mechanics program (Crutchfield and Hanson, 1997; Hanson and Crutchfield, 1992) builds local epsilon-machine models that identify domains, particles, and their interactions. Redeker (2017) formalized particle catalogs via de Bruijn diagrams. Rupe and Crutchfield (2018) generalized local causal states to arbitrary dimensions. Shalizi et al. (2006) developed automatic coherent-structure filters. These methods build rich local models at significant computational cost.

We take a different approach: rather than building local models, we ask a global model-selection question. *Given a finite family of relative-periodic templates, which template best describes the data?* We show that this question has a clean answer—the NML-optimal period stabilizes—and that the stabilization proof is dimension-agnostic, applying uniformly to 1D, 2D, and 3D automata.

Contributions

1. **Orbit-class reduction** (Theorem 2): fitting a relative-periodic background reduces to independent majority voting on orbit classes, yielding the Hamming-optimal projection in $O(|U|)$ time per candidate.

2. **Monotonicity** (Theorem 3): higher periods achieve lower residual counts along constant-velocity divisibility chains, necessitating complexity control.

3. **Model selection stabilization** (Theorem 5): under convergent orbit-class frequencies, every suboptimal candidate is eventually excluded; when the rate-minimizer is unique, selection locks.

4. **Nonidentifiability** (Theorem 7): background period recovery is impossible in general; periodic residual structure can be absorbed by higher-period templates.

5. **Cross-dimensional experiments**: empirical stabilization for 1D (ECA 30/54/110), 2D (773 totalistic + 106 LifeWiki rules), and 3D.

6. **Application**: identification of persistent structured projection residuals in 2D, with extensive scaling in S37/B11.

Relationship to Prior Work

Packard and Wolfram (1985) surveyed 2D rules qualitatively. Boccara and Roger (1999) identified period-2 families. Zenil (2010) used compression for CA classification. Our contribution is a *stabilization theorem* for period selection that applies across dimensions. Unlike computational mechanics (Crutchfield and Hanson, 1997; Rupe and Crutchfield, 2018), our model class is global (relative-periodic) rather than local (epsilon-machine), with the trade-off of missing local particle structure but gaining a provable stabilization guarantee and $O(|U|)$ cost per candidate.

Theory

Relative-Periodic Model Class

Definition 1 (Relative-periodic model class)**.** Given a binary spacetime $U \in \{0,1\}^{T\times D_1\times\cdots\times D_n}$, shift $\mathbf{s} \in \mathbb{Z}^n$, and period $p \geq 1$, the *relative-periodic model class* $\mathcal{B}(p,\mathbf{s})$ consists of all fields B satisfying $B[t+p,\ \mathbf{x}+\mathbf{s} \bmod \mathbf{D}] = B[t,\mathbf{x}]$ for all valid indices. This partitions spacetime into *orbit classes* $\{O_j\}_{j=1}^k$ where $k = p\prod_i D_i$. Within each class, B is constant.

Orbit-Class Reduction

Theorem 2 (Optimal Hamming Projection)**.** *Let* $n_j^{(1)} = |\{(t,\mathbf{x}) \in O_j : U[t,\mathbf{x}] = 1\}|$ *and* $n_j^{(0)} = |O_j| - n_j^{(1)}$. *The Hamming distance* $d(U,B)$ *is minimized over* $\mathcal{B}(p,\mathbf{s})$ *by majority vote:* $b_j = [n_j^{(1)} > n_j^{(0)}]$. *The minimum residual count is* $\sum_j \min(n_j^{(0)}, n_j^{(1)})$. *The minimizer is unique iff no orbit class has* $n_j^{(0)} = n_j^{(1)}$.

Proof. The Hamming distance decomposes additively over orbit classes. Since B is constant on each O_j, choosing b_j independently minimizes each $d_j = \min(n_j^{(0)}, n_j^{(1)})$. □

This is the *orbit-class reduction*: the spatiotemporal fitting problem decomposes into k independent binary estimation problems, exact in any dimension.

Monotonicity and Overcapacity

Theorem 3 (Monotonicity)**.** *If* $(p_2,\mathbf{s}_2)$ *is a velocity-matched multiple of* $(p_1,\mathbf{s}_1)$*—i.e.,* $p_2 = m\cdot p_1$ *and* $s_2^{(i)} \equiv m\cdot s_1^{(i)} \pmod{D_i}$*—then the orbit partition under* $(p_2,\mathbf{s}_2)$ *refines that under* $(p_1,\mathbf{s}_1)$*, and* $d^*(p_2,\mathbf{s}_2) \leq d^*(p_1,\mathbf{s}_1)$.

Proof. The periodicity map τ_2 equals τ_1^m, so τ_2-orbits refine τ_1-orbits. Optimizing over a finer partition cannot increase total disagreement (Theorem 2). □

Along any constant-velocity chain, the residual count is non-increasing. This *overcapacity problem* motivates complexity-penalized selection.

Bernoulli NML Criterion

Each model $(p,\mathbf{s})$ decomposes the data into $k = p\prod_i D_i$ independent Bernoulli estimation problems (Theorem 2).

Definition 4 (Bernoulli NML score)**.**

$$\mathrm{NML}(p,\mathbf{s}) = \underbrace{\sum_{j=1}^{k} n_j\, H_b(\hat{\theta}_j)}_{\text{NLL (data fit)}} + \underbrace{\sum_{j=1}^{k} \tfrac{1}{2}\log_2 n_j}_{\text{parametric complexity}},$$

where $\hat{\theta}_j = n_j^{(1)}/n_j$ and H_b is binary entropy. The complexity term is the asymptotic NML parametric complexity for Bernoulli families (Grünwald, 2007; Shtarkov, 1987).

The NML score depends only on orbit-class statistics $(n_j, n_j^{(1)})$—it is independent of spatial arrangement and traversal order.

Stabilization Theorem

Theorem 5 (Stabilization)**.** *Let $\mathcal{C}$ be a finite candidate set. Assume convergent orbit-class frequencies: $\hat{\theta}_j(T) \to \theta_j^*$ as $T\to\infty$. Define the per-site NLL rate $\lambda_c = \frac{1}{k_c}\sum_j H_b(\theta_j^*)$, and let $\lambda^* = \min_c \lambda_c$. Then:*

(i) $\mathrm{NML}(c,T) = \lambda_c \cdot N(T) + \frac{k_c}{2}\log_2 T + o(N(T))$.

(ii) *Every candidate with $\lambda_c > \lambda^*$ is eventually excluded.*

(iii) *If λ^* is achieved uniquely, selection locks to a single winner.*

Proof. Under convergent frequencies, $\mathrm{NLL}(c,T) = \lambda_c N(T) + o(N(T))$ and complexity is $\frac{k_c}{2}\log_2 T + O(1)$. For $\lambda_{c_1} > \lambda_{c_2}$, the $\Theta(T)$ gap dominates $O(\log T)$ complexity, so c_1 is eventually beaten. Since $|\mathcal{C}|$ is finite, all suboptimal candidates are excluded. □

Corollary 6 (Dimension-agnostic)**.** *Theorem 5 makes no reference to spatial dimension.*

Nonidentifiability and Identifiability

Theorem 7 (Nonidentifiability)**.** *For any period p_0 and shift* $\mathbf{s}$*, there exist spacetimes for which no MDL-type criterion (with $O(\log T)$ penalty) selects p_0 for large T.*

Proof sketch. Construct U with residuals under $(p_0,\mathbf{s})$ that are themselves periodic with period q. The candidate $(q\cdot p_0, q\cdot \mathbf{s} \bmod \mathbf{D})$ absorbs the residuals, achieving $\lambda = 0$ vs $\lambda_{p_0} > 0$.

Theorem 8 (Identifiability for exact backgrounds)**.** *If all orbit classes under p_0 have $\theta_j^* \in \{0,1\}$ (exactly periodic background), then NML selects p_0 over all velocity-matched multiples for large T.*

Proof. Both candidates have $\lambda = 0$, but the multiple has $m\cdot k_{p_0}$ orbit classes vs k_{p_0}. The complexity difference $\frac{(m-1)k_{p_0}}{2}\log_2 T \to +\infty$ favors p_0. □

Theorems 7 and 8 are complementary: NML recovers the true period when the background is exactly periodic, and absorbs periodic residual structure into higher-period templates when it is not.

Cross-Dimensional Experiments

1D Elementary CA

ECA rules 30, 54, and 110 on a ring of width 192, scanning shifts ± 6, periods 1–10.

Table 1: NML-optimal decomposition ($T = 144$).

Rule	(s, p)	Defect	NLL	Cmplx	NML
54	$(0, 4)$	20.1%	17,477	1,985	19,462
110	$(0, 7)$	30.8%	22,842	2,931	25,774
30	$(5, 1)$	46.3%	27,489	688	28,177

The known complexity hierarchy Rule 54 $<$ Rule 110 $<$ Rule 30 (Crutchfield and Hanson, 1997; Redeker, 2017) is recovered.

Table 2: Period stabilization (1D).

Rule	T=50	100	200	400	800	Margin
ECA-54	4	4	4	4	4	1,993
ECA-110	7	7	7	7	7	18,378
ECA-30	1	1	1	1	1	598

All three rules show *immediate* stabilization from T=50 with growing margins, consistent with Theorem 5.

2D Totalistic Rules

Range-threshold survey. We survey 1,050 range-threshold totalistic rules on a 48×48 torus (Moore neighborhood, 40 steps, density 0.5, seed 11). Of 773 non-trivial rules, 614 select period 1, 159 select period 2, and 172 have nonzero best-fit shift.

LifeWiki named rules survey. We survey all 106 named Life-like rules from the LifeWiki, including rules with non-contiguous B/S sets (e.g., B36/S23, B3678/S34678), on a 64×64 torus with periods 1–8 and shifts ± 2.

Four rules change period between T=100 and T=400 (two additional rules became trivial, accounting for the non-trivial count dropping from 105 to 103):

- **Diamoeba** (B35678/S5678): p=1 $\rightarrow$ p=2. NLL gap grows from 7,749 to 17,085 bits, crossing the complexity threshold at $T \approx 400$.
- **Maze with Mice** (B37/S12345): p=1 $\rightarrow$ p=2.
- **B017/S01**: p=2 $\rightarrow$ p=4. Deeper periodic structure revealed.

Table 3: LifeWiki survey period distribution (nontrivial rules only; 1 trivial at T=100, 3 at T=400).

Period	T=100	T=400
$p = 1$	97	94
$p = 2$	7	7
$p = 4$	0	1
$p = 8$	1	1
Nontrivial	105 / 106	103 / 106

- **Serviettes** (B234/S): p=2 $\rightarrow$ p=1. Transient signal correctly eliminated.

This is consistent with Theorem 5: genuine periodic signals grow as $\Theta(T)$ and overcome the $O(\log T)$ complexity penalty, while spurious signals are eliminated.

Table 4: LifeWiki rules with period > 1 at T=400.

Rule	Name	p	Margin
B01245/S01245	—	2	23,437
B014/S2	Oils	2	24,354
B026/S1	—	2	7,441
B345/S5	Long Life	2	23,673
B35678/S5678	Diamoeba	2	369
B37/S12345	Maze w/ Mice	2	10,230
B45678/S5678	Majority	2	25,169
B017/S01	—	4	3,106
B1357/S02468	Fredkin	8	153,305

Fredkin (B1357/S02468) achieves period 8—the only period > 4 in the catalog—due to its parity-rule replication cycle. Diamoeba's small margin (369 bits) reflects its recent transition; margins grow as Theorem 5 predicts.

Table 5: Period stabilization (2D).

Rule	T=60	100	200	400	800	Margin
S24/B11	1	1	1	2	2	13,115
S11/B37	2	2	2	4	4	22,352
S37/B11	1	1	2	2	2	16,006

Period stabilization (2D). These rules show *progressive* stabilization: the period increases in jumps then locks with growing margins. This is the regime of Theorem 7: periodic residual structure is absorbed into higher-period templates. Notably, NML selects *lower* periods than a previous RL-based MDL formulation (e.g., period 2 vs period 6 for S24/B11), because the RL artifact inflated data-fit terms for higher periods.

Multi-seed robustness. We verified 6 rules across 10 seeds (64×64, 60 steps):

Table 6: Multi-seed robustness.

Rule	p	Rate	CV	Cons.
S14/B11	1	0.59%	8.4%	10/10
S25/B12	2	0.58%	15.6%	8/10
S66/B36	2	0.63%	10.5%	10/10
S24/B11	2	2.78%	9.4%	10/10
S11/B37	4	2.75%	18.7%	7/10
S37/B11	2	4.41%	12.4%	10/10

Rules with near-zero NML margins (S25/B12, S11/B37) show seed-dependent period selection at short horizons—consistent with Theorem 5, which requires the linear NLL gap to dominate at large T.

3D Totalistic Rules

Diamoeba3d (survive 5–8, birth 5–8) on a 16^3 torus:

Table 7: Period stabilization (3D).

T	Period	NML Bits	Margin
10	6	26,406	5,560
20	6	75,703	3,055
40	1	158,154	3,505
60	1	232,856	4,872
80	1	303,209	5,840

The initial period 6 selection is a small-sample artifact (1–3 cycles per orbit class). By $T{=}40$, NML stabilizes to period 1 with growing margins.

Stabilization Summary

Table 8: Cross-dimensional stabilization.

Dim	Rules	Onset	Behavior
1D	ECA 30/54/110	$T{=}50$	Immediate
2D	3 pers.-res.	$T{=}200$–400	Progressive
2D	106 LifeWiki	$T{=}100$–400	4 change; 103 stable
3D	diamoeba3d	$T{=}40$	Transient, locked

Application: Persistent Structured Defects in 2D

400-Step Verification

All three rules (S24/B11, S11/B37, S37/B11) show stable defect populations from $t{=}100$ through $t{=}400$ on a 64×64 grid, with mean defect counts varying by $< 15\%$ between early and late windows.

Table 9: S37/B11 size scaling (extensive).

Grid	Mean Rate	Late Defects	Density
32^2	3.47%	11.4 ± 5.3	0.011
64^2	3.26%	45.7 ± 6.6	0.011
96^2	3.11%	101.9 ± 8.0	0.011
128^2	3.33%	196.5 ± 10.6	0.012
192^2	3.14%	428.5 ± 32.4	0.012

Size Scaling

Five seeds, grid sizes 32–192, 100 steps. *Defect density* is late defects divided by spatial grid area D^2.

Defect density is approximately constant ($\sim$0.011) across all grid sizes, consistent with a bulk (extensive) phenomenon.

Table 10: S11/B37 size scaling (sub-extensive).

Grid	Mean Rate	Late Defects	Density
32^2	1.46%	1.6 ± 0.6	0.0016
64^2	1.81%	11.1 ± 6.3	0.0027
96^2	1.71%	18.6 ± 6.2	0.0020
128^2	1.58%	32.7 ± 4.1	0.0020
192^2	1.69%	79.5 ± 16.7	0.0022

S11/B37 defect density drops then stabilizes at $\sim$0.002, consistent with sub-extensive scaling.

Discussion

What the Theory Provides

The orbit-class reduction transforms spatiotemporal fitting into standard Bernoulli estimation, enabling: (1) Hamming-optimal decomposition in $O(|U|)$ time for any dimension; (2) a formal explanation of overcapacity (Theorem 3); (3) a stabilization theorem (Theorem 5) with no assumption on residual geometry; (4) a nonidentifiability result (Theorem 7) with recovery guaranteed when the background is exactly periodic (Theorem 8).

The same framework—orbit classes $\rightarrow$ Bernoulli estimation $\rightarrow$ NML stabilization—applies without modification across spatial dimensions.

NML vs Legacy MDL

A previous formulation used run-length codelength as the data-fit term. This had three weaknesses: model mismatch (RL is not the Bernoulli NLL), geometry dependence (RL depends on traversal order), and an extra convergence assumption. NML and MDL agree for 1D rules but diverge for 2D rules with periodic residuals, where NML selects lower, more parsimonious periods (e.g., period 2 vs period 6 for S24/B11).

Limitations

Monotonicity requires velocity matching (satisfied for shift-zero, the dominant case). The Bernoulli i.i.d. assumption within orbit classes ignores temporal correlations (which do not affect asymptotic rates). NML selects the best-compressing period, not necessarily a physicist's "true" period (Theorem 7). The method identifies *where* residuals are but not *how* they interact—it does not recover domain/particle structure in the computational mechanics sense (Crutchfield and Hanson, 1997).

Future Work

Component tracking (lifetimes, velocities, collision catalogs), direct comparison with local causal states (Rupe and Crutchfield, 2018), larger surveys (1000+ 2D rules), exact finite-sample NML normalizers, and joint shift–period optimization.

Conclusion

We have proved that Bernoulli NML model selection over relative-periodic CA backgrounds stabilizes: every candidate whose per-site NLL rate exceeds the minimum is eventually excluded (Theorem 5). When the rate-minimizing candidate is unique—as in all cases we test—the selected model locks, though the selected period need not correspond to any external notion of "true background period" (Theorem 7). The proof requires only convergent orbit-class frequencies, is dimension-agnostic, and rests on the orbit-class reduction that decomposes spatiotemporal fitting into independent Bernoulli estimation.

Cross-dimensional experiments on 1D elementary CA, 2D totalistic and Life-like rules (including all 106 named LifeWiki rules), and a 3D totalistic rule are consistent with the stabilization prediction. As an application, the framework identifies 2D rules with persistent structured projection residuals exhibiting extensive scaling.

Reproducibility

All code is open-source. Key parameters: 1D ECA: width $= 192$, steps $= 144$, density $= 0.5$, seed $= 11$, shifts $= -6..+6$, periods $= 1..10$. 1D convergence: $T \in \{50, 100, 200, 400, 600, 800\}$. 2D range-threshold survey: 48×48, steps $= 40$, 1,050 candidates (773 nontrivial). 2D LifeWiki survey: 64×64, steps $= \{100, 400\}$, density $= 0.5$, seed $= 42$, periods $= 1..8$, shifts $= \pm 2$. 2D convergence: 64×64, $T \in \{60, 100, 200, 400, 600, 800\}$. 3D: 16^3, $T \in \{10, 20, 40, 60, 80\}$. Multi-seed: seeds $= \{11, 42, 73, 99, 137, 200, 314, 500, 777, 1024\}$. Size scaling: grids $\in \{32, 64, 96, 128, 192\}$, steps $= 100$, 5 seeds. 400-step verification: 64×64, seed $= 11$.

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